

Index laws



1

a 2^{21}

e 5^{48}

i 19^5

b 22^8

f 15^4

j 12^7

c 11^4

g 23^{79}

k 13^8

d 7^5

h 17^8

l 15^{27}

2

a 8

e 3

i 5

b 7

f 6

j 2

c 34

g 5

d 5

h 38

3

a 1 679 616

e 16 807

i 36

b 9 261

f 4 096

j 1

c 1 024

g 121

k 1 728

d 390 625

h 729

l 36

Negative bases

4

a $(-11)^{13}$

e -12

i $(-7)^5$

b $(-21)^8$

f $(-30)^3$

j $(-6)^2$

c -15^2

g 8^3

d $(-3)^7$

h 11^{11}

5

a -7

e 49

i 3

b -3

f 12

j -55

c 50

g -2

d 2

h -21

6

a 256

e -27

i 1 225

b 64

f -3 125

j 531 441

c -243

g -216

k 64

d -8 000

h -1 977 326 743

l -8 000



Fractional bases

7

a $\frac{1}{3^4}$

b $\frac{3^3}{8^3}$

c $\frac{1^8}{4^8}$

d $\frac{5^2}{2^2}$

e $\frac{10^5}{33^5}$

f $\frac{2^6}{35^6}$

g $\frac{5^3}{18^3}$

h $\frac{29^7}{41^7}$

i $\frac{11^9}{13^9}$

j $\frac{20^2}{3^2}$

k $\frac{17^4}{4^4}$

l $\frac{31^5}{50^5}$

8

a 3

b 3

c 2

d 1

e 3

f 16

g 9

h 625

i 2

j 121

k 5

l 5

9

a $\frac{1}{9}$

b $\frac{27}{125}$

c $\frac{1}{32}$

d $\frac{49}{121}$

e $\frac{27}{512}$

f $\frac{144}{169}$

g $\frac{125}{729}$

h $\frac{16}{81}$

i $\frac{4\,096}{15\,625}$

j $\frac{8}{125}$

k $\frac{25}{49}$

l $\frac{1\,000}{2\,744}$



10

a No

b Yes

c No

d Yes

Negative indices



11 Less than 1.

12

a $\frac{1}{a^9}$

b a^n

c $\frac{1}{4a^9}$

d $\frac{b^m}{a^n}$

e $\frac{1}{p^2}$

f $\frac{3}{x^4}$

g $\frac{7}{x^9}$

h $\frac{q^3}{p^2}$

i $\frac{8}{p^3}$

j $\frac{2y^3}{x^8}$

k $\frac{c^3}{6b^7}$

l $\frac{p^4}{m^5n^4}$

m $\frac{3a^4}{b^5c^7}$

n $\frac{l^7}{16k^3}$

o $\frac{g^2}{f^7h^8}$

p $\frac{s^4u^6}{5t^9}$

13

a x^{-4}

b $2x^{-6}$

c xy^{-2}

d $4x^{-6}a^{-7}$



14

a $20y^6$

b $28a^2$

c $\frac{-15}{x^4}$

d $\frac{12}{y^5}$

e $8h^7$

f $\frac{6}{y^7}$

g $\frac{16}{y^3}$

h $25m^3$

i $\frac{3}{x^7}$

j $\frac{3}{x^4}$

k $3x^8$

l $\frac{3}{2c^2}$

m $\frac{5h^9}{4}$

n $3x^8$

o $\frac{3}{2p}$

p $\frac{9}{4w^{10}}$

15

a $\frac{1}{8m^3}$

b $\frac{256}{m^{24}}$

c $\frac{p^8}{9}$

d $\frac{1}{9y^4}$

e $\frac{64}{y^3}$

f $\frac{y^4}{x^5}$

g $\frac{y^{36}}{x^{28}}$

h $5x^5$

i $\frac{5}{x^4}$

j $\frac{81}{z^4}$

k $\frac{q^7}{p^3}$

l $\frac{x^8}{y^{16}}$

16

a $n = -2$

b $n = -3$

c $n = -4$

d $n = -3$

17 $(x^4y^{-3})^{-4} = \frac{y^{12}}{x^{16}}$

18 The -1 index only applies to the term directly in front of it. In this case it only applies to the a , not the 5. The correct working is $5a^{-1} = \frac{5}{a}$.

19

a $32y^8$

b $60y^5$

20

a $\frac{10z^9}{21x^3y^7}$

b $\frac{160y^6}{27x}$